

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26136

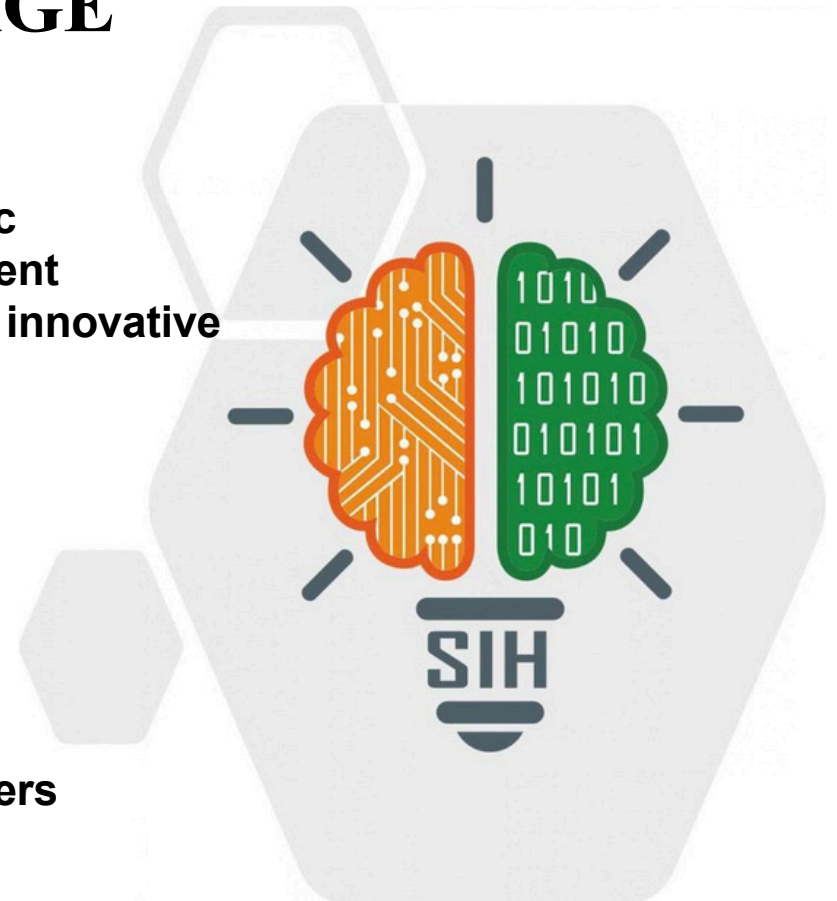
Problem Statement Title- Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups

Theme- Miscellaneous

PS Category- Software

Team ID - MITS/SIH26/191

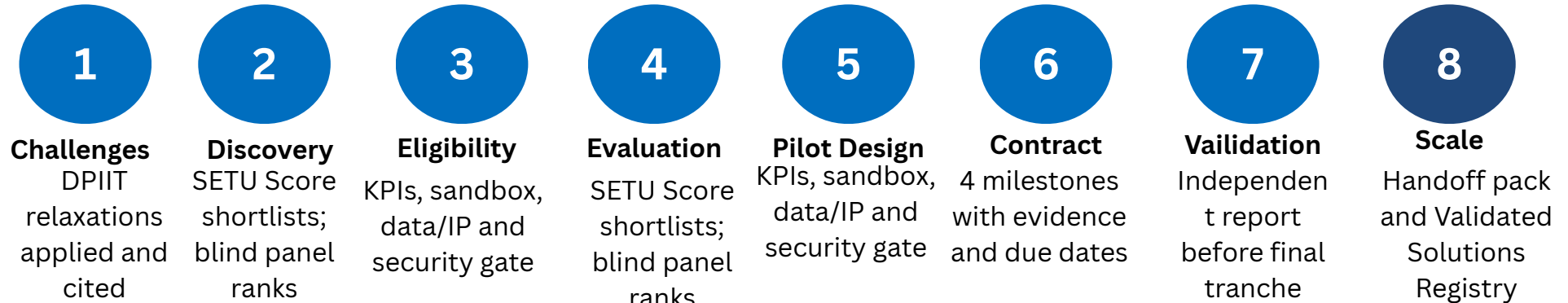
Team Name (Registered on portal) – The prisoners





Our solution : one case, eighth gated stages

Our strategy: every step is rule-driven, explainable and logged, and no stage can be skipped. SETU prepares and evidences; purchase and payment stay in GeM, CPPP and PFMS.



WHAT OUR PROTOTYPE DOES TODAY

Conflict check

Warns or blocks a startup already in an active pilot

Smarter matching

Synonyms such as traffic and congestion, with reasons

Officer verification

Pending to verified after manual review.

Department analytics

Problem posted, active pilots, budget disbursed

What it is built with

Backend – Python + FastAPI, Pydantic validation

Database – SQLAlchemy + Alembic; SQLite
(**prototype**) → PostgreSQL (production)

Access – Role-based: Department, Startup, Expert, Validator, Admin

Matching – TF-IDF **today**; SBERT embeddings in v2 – CPU-only, no paid AI API, no GPU

Eligibility – Rules as versioned data; DPIIT relaxations applied with citation

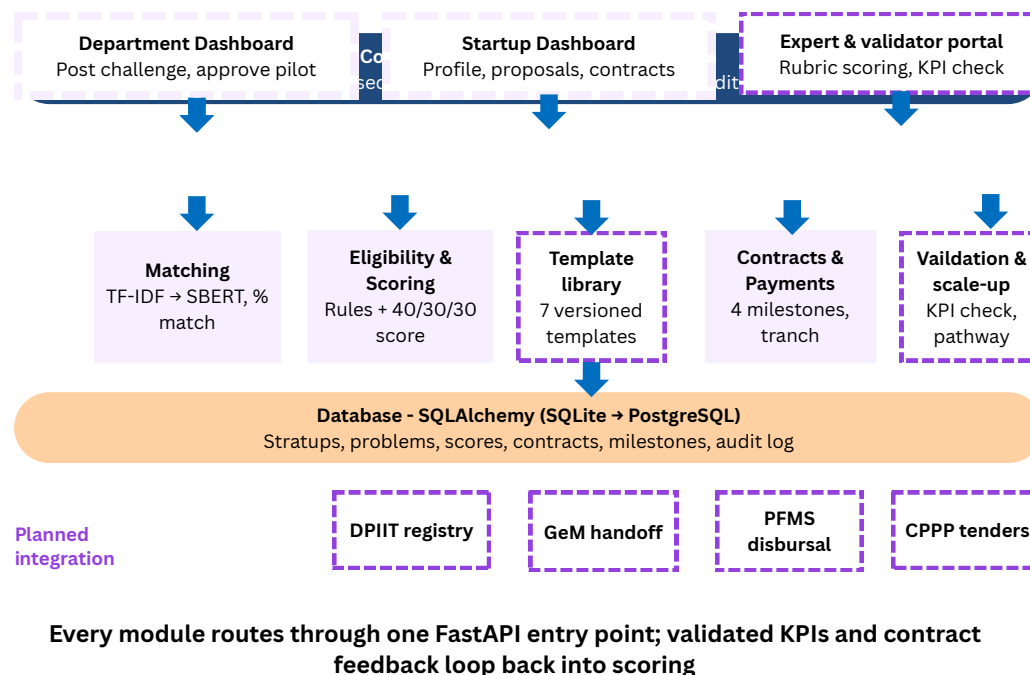
Scoring – 40% Fit + 30% Profile + 30% Feedback; score shortlists, panel decides

State machine – Gated stages for challenge, application, milestone – none skippable

Audit – Append-only, hash-chained log; tamper-evident

Frontend – HTML + CSS + JavaScript dashboards per role

Process flow architecture



How a challenge becomes a scaled solution

1. Post outcome-based challenge (baseline, KPI, budget)

2. Matching engine ranks startups with % match + reason

3. Eligibility checked; technical capability never relaxed

4. Expert panel scores blind on a published rubric

5. Pilot plan + agreement (data/IP, cybersecurity, risk)

6. Four-milestone contract: evidence → verify → pay

7. Independent validator file KPI report before final tranche

8. Scale decision; handoff pack + registry for other depts

Feasible today, viable at scale

68% of GeM orders in FY 2025-26 went to micro and small enterprises (47.1% of GMV). Small suppliers already sell to government at scale

Why it is Feasible:

- Legal basis exists: GFR 2017 Rules 170(i) and 173(i)
- Local supply: 35,992 recognised startups in Maharashtra
- Open-source stack, no GPU, one node
- Adapters keep the demo independent of outside access

Risks and how we handle them

RISK	MITIGATION
Rules misread procurement law	Versioned rule sets, officer approval, citations shown
Officers see it as extra work	Handoff pack + GeM-compatible export, no double entry
Score gaming	Verified evidence outweighs claims; expert panel decides
Reviewer bias	Blind scoring, conflict declarations, spread flags, audit trail
No real DPIIT/GeM access yet	Adapters degrade to “Manual review”; demo never blocks

ROLLOUT

Pilot

One Maharashtra department, one live challenge

Scale

More departments and districts; SBERT matching if it beats keywords

Pilot

Real DPIIT lookup, GeM export and PFMS reference

Who gains, and how we prove it

Payment delays lock about ₹8.1 lakh crore of MSME money. SETU makes every due date visible.

1 Departments

- A ranked, defensible shortlist once applications close
- Controlled pilots with a security and data gate
- Independent validation before any scale decision

2 Stratups

- Public board shows real departmental demand
- Fair screening: turnover and experience relaxed, capability never
- Payment tied to verified milestones with a visible due date

3 Maharashtra

- Validate once, replicate across departments and districts
- Every decision reconstructable from the audit trail
- A bridge for 35,992 state startups to reach public buyers

HOW WE WILL PROVE IT: TARGETS ARE SET WITH THE PILOT DEPARTMENT AFTER A BASELINE

Time to shortlist

Minutes from applications closing to shortlist

Outcome quality

% of challenges published with a KPI, baseline and target

Payment timeliness

% of milestone paid by the due date

Evidence

% of pilots with an independent vailadation report

Conversion

% of vailadated pilots scaled, and replications

Policy and ecosystem research

- **GFR 2017 Rule 173(i)** – relaxes prior turnover and experience for DPIIT-recognised startups; technical/quality specs never relaxed
- **GFR 2017 Rule 170(i) (amended 2017)** – exempts DPIIT startups and MSEs from bid security/EMD
- **GeM Startup Runway** – closest existing mechanism: 15-day product / 8–16 week service trial, needs 3+ ratings from 3+ buyers before listing; no expert panel, milestone payments, or independent validation – the gap SETU fills
- **CPPP (eprocure.gov.in)** – central tender portal; same DPIIT relaxations apply
- **PFMS** – public fund tracking and disbursement
- **MoD iDEX / Make-II** – problem-led, collaborative Indian precedent from another ministry
- **NITI Aayog, Maharashtra Startup Week** – state-level government–startup engagement

International comparators

- **UK SBRI / Contracts for Innovation** – problem-first competitions, department as "lead customer," pre-commercial risk-sharing, IP stays with supplier
- **US SBIR** – the older US model SBRI was built on
- SETU borrows the challenge-led, staged-risk approach but structures pilots as milestone contracts inside GFR/GeM rules, not open-ended R&D grants

Research insight

Existing procurement and startup ecosystems (DPIIT, GeM, CPPP, PFMS) already give access to buyers, sellers and pilot opportunities – but no single mechanism carries a case through **problem → discovery → screening → evaluation → pilot → validation → payment → scale**. SETU joins these into one auditable pathway.